



Digit: The Bonus Thumb

A Report Prepared For:

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Dear Spin Master team,

This report, titled Digit: The Bonus Thumb, was prepared for the ME100 final toy design project. It presents our toy concept and prototype along with an analysis of the toy's mechanical feasibility and play value for children.

The primary purpose of this report is to present the team's evaluation of Digit's mechanical feasibility. The report presents the team's functional breakdown, constraints and criteria, solutions developed, and the design decisions made. It will also demonstrate the mechanical engineering concepts used to prototype Digit.

Below is a summary of the contributions from each person to the project:

J.B.	• Thumb and wrist actuation design and CAD • Isolated thumb yaw and pitch actuations • Integration of sub-components in CAD
S.E.	• Thumb design and CAD • Joint manipulation
Trevor Werbin	• Mounting thumb to hand • Cap to put on thumb to attach string • Designing and CAD for elastic
J.G.	• Pulley adjustment system (to account for varying hand sizes) • Yaw axis assembly
M.U.	• Worked with Trevor on thumb cover • Working on various designs for thumb actuation • Designing and CAD for elastic

The team hopes that this report and viability assessments will be taken as substantive recommendations.

This report was written entirely by the team and has not received any previous academic credit at this or any other institution.

Best Regards,

J.B.

S.E.

Trevor Werbin

J.G.

M.U.

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Summary

Digit is a mechanically actuated wearable toy that provides children with an additional thumb to enhance their grasping ability. It is designed to be powered by the body and to be electronics-free. Digit demonstrates how an additional thumb concept can be used in a mechanically viable prototype that aligns with Spin Master's goals for safe, engaging and manufacturable toys.

The report begins by highlighting Digit's appeal for children aged approximately 6–12, noting that it enables them to experiment with grasping multiple items in one hand. It is argued that Digit provides many diverse and stimulating play patterns. Comparable toys to Digit include robotic hands, mechanical gripper toys and products inspired by prosthetics and exoskeletons.

The primary challenge identified during prototyping was the absence of an affordable, safe, and easy-to-learn wearable toy that gives children an extra thumb to enhance grip, interaction with toys, and imaginative role-play of superhuman abilities. From this problem, the key functions were defined, including securely mounting Digit to the hand and achieving a sufficient range of motion. These functions are tied to specific constraints and criteria, including target values for range of motion and gripping force.

The report outlines how to address the required functions and provides a structured exploration of the design solutions considered for the five primary mechanisms: mounting on any size hand, actuating the thumb, retracting the thumb, extending the thumb and incorporating a yaw mechanism that operates independently of the other degrees of freedom.

To retract and extend Digit, there were two analyzed approaches: a thumb-operated winch and a wrist-driven linkage. The thumb-based concept was used for the first prototype but proved counterproductive, as it required sacrificing the use of the user's own thumb. The current prototype adopts the wrist-driven linkage, which has improved performance by using natural wrist motion to extend and retract Digit.

The technical progress sections discuss the various tools and technologies used to create an integrated prototype through 3D printing and utilizing pins and bearings. The linkages and joint placements were iteratively refined via SolidWorks, and manual testing of Digit measured approximately 90 degrees of motion for the thumb retraction and around 120 degrees of movement for the yaw.

The report ends with identifying several remaining challenges, but the overall conclusion is that Digit is mechanically viable, as proven through the prototyping, and that Spin Master should continue development of the product.

1.0 Introduction

This section introduces the overall concept of Digit, including its play pattern, comparable products, and potential applications. It also discusses the inspiration behind Digit and how the prototype has been customized to make it unique.

1.1 Play Pattern / Age Range

Digit allows children aged 6-12 to explore a variety of play patterns. Firstly, it provides additional utility for grasping items and carrying more of them as the extra thumb can actuate independently of the other fingers. This is seen in Fig. 1 where Digit is retracted inward while the rest of the fingers are at their resting position. This solution emerged from the iterative process of determining how to produce a purely mechanical action using some movement of the hand or wrist. Furthermore, Digit provides an educational opportunity for children to learn about the physics of motion and how one action can inadvertently cause another in an interesting way. Additionally, Digit teaches children about concepts in manipulation and control systems, as it enables them to accomplish the task of picking up an object by rotating and articulating the toy through wrist movement.

1.2 Comparable Products

Digit draws on a diverse range of products and uniquely integrates elements from existing toy designs. Some technologies adjacent to Digit include mechanical gripper toys, robotic hands, prosthetics, etc. As seen in Fig. 2, which is a toy similar in function to Digit, mechanical grippers typically employ linkages connected to a rod to actuate a set of claws to pick up an object [1]. Similarly, as shown in Fig. 1, Digit employs a linkage to translate rotational wrist movement into the rotation of the additional thumb. Robotic hand toys commonly employ tension-based actuation using ropes and elastic bands, translating pressure on a switch or finger movement into curling, extension, or rotation of fingers [1]. Although Digit relies on ropes and elastics to extend and curl the thumb, the action that produces the movement primarily comes from the wrist, as opposed to the fingers. This feature enables improved mobility while preserving the function and form of the toy. Finally, prosthetics and exoskeletal devices serve both functional and recreational purposes. While prosthetics are primarily designed to replace missing or impaired limbs, Digit, though not intended for this purpose, lends itself to the idea of improving the functions and abilities of the human body by providing an additional thumb.



Fig. 1: Final prototype of Digit retracted



Fig. 2: Robot grab hand comparable product [1]

1.3 Objective

The objective of the report is to demonstrate the mechanical feasibility of Digit, identify remaining challenges, and outline future development opportunities to inform Spin Master on whether to pursue this toy.

2.0 Problem Definition

Children between the ages of 6–12 are drawn to toys that allow them to imagine, experiment, and demonstrate new abilities. In contrast, existing wearable robotic devices are often designed for adults, expensive, and primarily serve industrial or medical purposes. These devices typically require a long learning period before they can be used effectively and often demand fine motor-control skills that children have not yet developed.

Current “extra limb” or robotic-thumb devices are typically too heavy, difficult to control, or unsafe for young users. As a result, children do not have access to a playful, lightweight, and intuitive wearable device that lets them experience controlling a non-natural appendage through trial-and-error learning.

An extra thumb also has the potential to expand how children play. From building LEGO structures to manipulating small objects like marbles, an extra thumb can introduce entirely new play styles by enhancing grip, precision, and the variety of ways toys can be handled.

The core problem, therefore, is the lack of an affordable, safe, and easy-to-learn wearable toy that provides children with a functional third thumb to enhance grip, interaction with toys, and imaginative role-play of superhuman abilities.

2.1 Required Functions

Digit is designed to be simply, predictably, and intuitively controlled. A child should be able to understand how the thumb will move based on their natural wrist and thumb motions without requiring complex training. The device should also be adjustable to different hand sizes, since children between the ages of 6–12 grow quickly and vary widely in hand dimensions.

The toy is aimed to be durable, flexible, and safe for repeated play. The extra thumb will be capable of picking up objects using the pivot of the wrist and allowing items to rest securely against the palm. To enable the most variable play possibilities, the movements of the extra thumb must be smooth, reliable, and versatile, including the ability to rotate, grasp, cling, and reposition as needed. These capabilities will allow the child to explore new styles of interaction and develop new play mechanics that are not possible with only a biological hand.

2.2 Constraints and Criteria

Table 1 presents five different ranked functions and their respective constraints and criteria, each selected for their significant impact on the design, safety, and overall functionality of Digit. The highest ranked function is that Digit must remain securely mounted on the user’s hand throughout motion and actuation of the finger. This function is fundamental to maintaining user safety, further discussed in Section 2.3, and device stability, as any slipping or unintended movement of the mount would make the toy very difficult to use effectively. The constraints for this function require Digit to accommodate the 5th to 95th percentile of hand sizes within the toy’s target age range. This function is ranked as the most critical to achieve since the other four functions, which are mainly movement related, are dependent on secure attachment and correct mounting orientation. Without it, Digit will not operate as intended.

The remaining functions address the movement capabilities of Digit which include actuation, retraction, extension, and yawing functions. The constraints ensure that Digit can provide sufficient and consistent movement of the finger across the x, y, and z axis to mimic how a human thumb moves as accurately as possible. Each constraint covers properties such as actuation speed, load bearing capacity and angular range and contain criteria, such as having as low resistance as possible during extension of Digit, helping to ensure that the toy will perform reliably across multiple scenarios and motions. The values for the constraints were chosen to replicate children's finger strength [2].

Table 1: Digit Functions, Constraints, and Criteria

	Function 1	Function 2	Function 3	Function 4	Function 5
Prioritized Functions	Mount on Hand	Actuatable Thumb	Retract Thumb	Extend Thumb	Yaw Thumb
Constraints	- Stay mounted during motion and actuation - Mount on 5-95 percentile class hand sizes	>80 degrees range of motion - Withstand >1000 actuation fatigue cycle - String tensile strength <45N	- Hold >250 grams - Controlled independently of yaw - Forward retraction/rotation >80 degrees	- Actuate <1s - Controlled independently of yaw - Backwards extension/rotation >80 degrees	- Controlled independently of retraction/extension >80 degrees range of motion
Criteria	- Least pressure on hand - Least sliding during motion	Lowest friction during motion	- Least stress on actuating muscles - Most dexterity during actuation	- Lowest resistance during retracting - Most repeatable movement	- Lowest stress on actuation muscles - Lowest stress holding yaw position - Closest mounting to hand

2.3 Safety Precautions

Digit is attached to the hand using a glove-like component so that it remains securely in place. The finger screws are fully enclosed within the 3D-printed structure, preventing any sharp edges

from being exposed or coming into direct contact with the skin. The string-tensioning fasteners are tightened using locknuts, which require tooling to remove, largely preventing choking hazards. All potentially sharp design edges have been smoothed with fillets.

When placing the extra thumb on the hand, it mounts with hook and loop strapping, holding it securely without the risk of over-tightening. This ensures long-term use of the glove will not cause itching or discomfort. The glove piece is made as a single, solid component, meaning it has no small parts that could break off, be picked up, or pose a choking hazard.

The wrist movement was designed to be actuated with cable ~300mm long, satisfying the <500mm regulation determined by Health Canada [3].

3.0 Technical Progress

This section details the research, design details, and viable solutions developed to address the functional requirements for Digit, which are outlined in Section 2, as well as the technical progression to date. Digit's design is centered on translating human wrist movements into independent movements on the extra thumb using four primary mechanisms: linkage-based yaw, single joint finger design, a cable driven retraction system and an elastic extension mechanism. Through iterative development, mechanical exploration, and targeted engineering decisions, the current prototype of Digit meets all of the design constraints and demonstrates viability towards further development.

3.1 Solutions Considered

Multiple solutions were considered for the four primary functions of Digit, defined in Section 2.3: mechanical thumb that is actuatable, yawing the thumb, retracting the thumb, and extending the thumb.

3.1.1 Mechanical Thumb

Design constraints for the mechanical finger were achieving a fingertip range of motion exceeding 80 degrees and withstanding 1000 actuations. Because hinge performance is critical to finger functionality, it became the primary focus of the research. Two hinge concepts were evaluated: a living hinge, in which all three joints are integrated into a single component, and a roll-pin hinge where each joint of the finger is separate. The living hinge offers simplicity and improved safety by eliminating small detachable parts that could pose choking hazards. Consequently, a living hinge was selected for initial testing, with the roll-pin hinge being tested after. For further details on hinge design and solutions, see Section 3.2.1.

3.1.2 Finger Yawing

Constraints for yawing the thumb were >80 degrees of movement and <1 second actuation time. Two main solutions were evaluated: a string-based winched and a rigid linkage. The string-based design uses the yawing motion of the wearer's wrist to pull on a string, causing the thumb to yaw. A torsion spring would apply tension to the string, allowing the thumb to rotate in both directions.

The string-based design seen in Fig. 3. has two major advantages to the linkage-based design. The string compliancy removes strain on the wearer's joints, as it does not force the wrist to follow a rigid path like the linkage design. The string-based design is also more durable, as shock loads from falling are less likely to break a compliant mechanism.

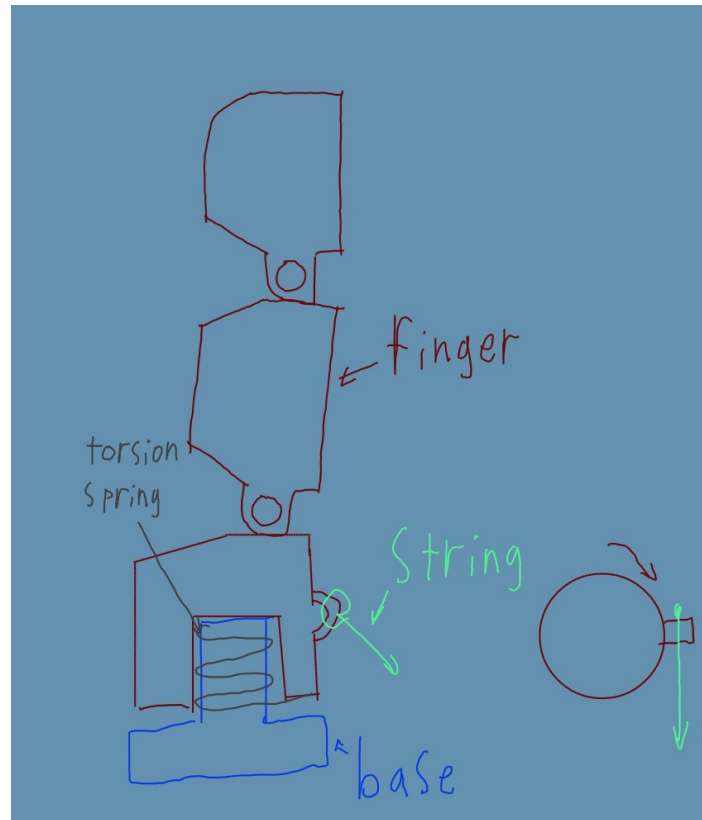


Fig. 3: Sketch of string-based yaw mechanism

The linkage-based design has two major advantages over the string-based approach. First, by converting wrist yaw into bidirectional thumb rotation, the mechanism provides active control in both directions, unlike the string-based design where one load direction relies solely on spring force. This limitation in the string-based system reduces load capacity unless a high-stiffness spring is used, which would increase actuation effort and reduce useability. Second, the linkage-based design eliminates the need for constant holding force, mitigating user fatigue associated with the constant tension required in rope-and-spring systems.

Since the play pattern is designed for indoors, shock loads are a small concern. Both designs satisfy the constraint of <1 s actuation time. However, the linkage design satisfied the criteria of least stress on actuation muscles. Therefore, the linkage design was chosen.

3.1.3 Thumb Retraction

Retracting the thumb had three constraints: actuate from human movement, transmit >10 N of force, and <1 s actuation time. The research demonstrated that string-based designs are the most common, so it was the focus of ideation. Within string-based design, actuation through the wearer's thumb, and actuation through the wearer's wrist were both considered. Digit uses a

similar mechanism to the one shown in Fig. 4, which uses the angling of the finger to change string path length, retracting the rest of the prosthetic.

Actuation through the wearer's thumb was investigated first as the entire retraction mechanism moves together during yawing, avoiding path-length changes. However, requiring the wearer's thumb to move simultaneously limits dexterity. Consequently, wrist-based actuation was selected for the final prototype to maximize functional independence and user control.

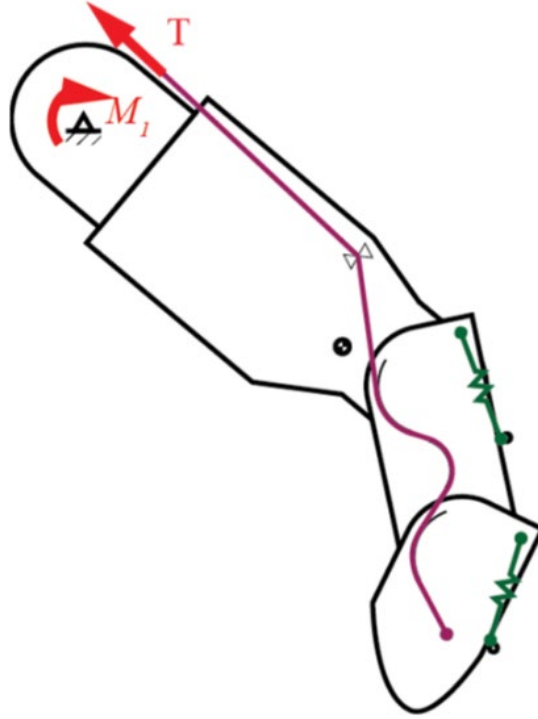


Fig. 4: Illustration of finger actuation from the wearer's wrist [4]

3.1.4 Thumb Extension

Extending the thumb had two constraints: >80 deg range of motion and <1 s actuation time. Two solutions were looked at: TPU joints as shown in Fig. 5, and elastic band retraction. TPU joints simplify part count, as their natural elasticity allows the TPU to function both as the thumb joint and as the mechanism to extend the thumb back to the straight position. While both solutions satisfy the <1 s extension constraint, elastic bands were selected because they allow rapid tuning of spring force, whereas the TPU joints require slower adjustments through 3d-printed geometry changes.

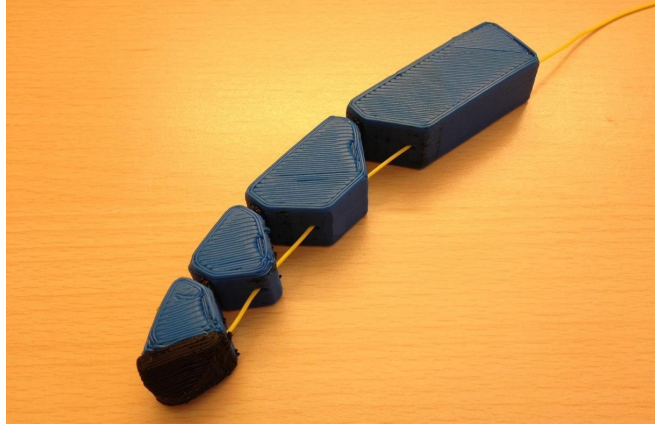


Fig. 5: A TPU finger joint [5]

3.2 Progress to Date

Fig. 6 shows a retrospective Gantt chart of the project's design timeline. Parallel development was employed for as many functions as possible to enable rapid iteration, but highly integrated mechanisms such as the yawing linkage required initial design of other components to satisfy the closest mounting point criteria.

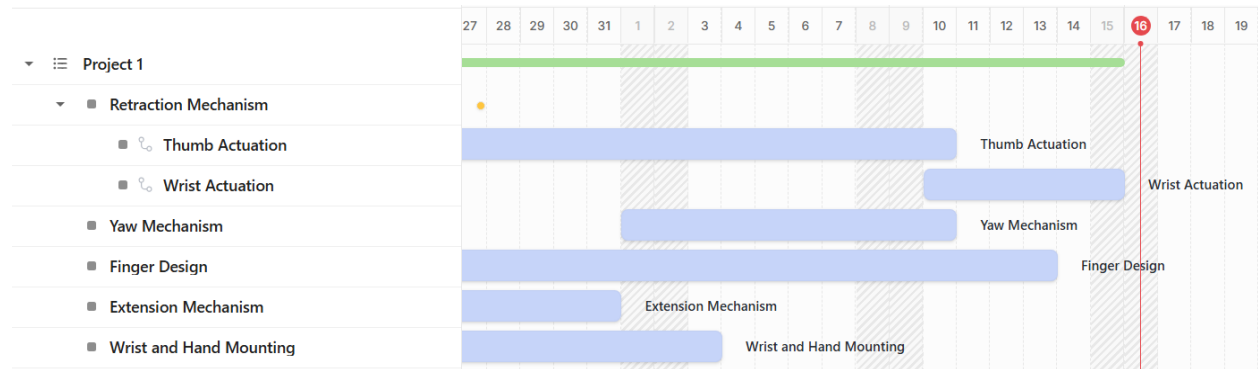


Fig. 6: Gantt chart of the design process

3.2.1 Thumb Hinge Design

Digit comprises three articulated thumb segments that must rotate independently to replicate natural thumb motion, necessitating a hinge mechanism capable of sustaining repeated flexion under load. The hinge also had to minimize spatial footprint to accommodate an elastic band channel behind the thumb while resisting backward peeling forces from the return. Two hinge types were evaluated to meet these requirements: living hinges and revolute joints. All components were printed in PETG, selected for its improved toughness over PLA, though thin geometries still pose fatigue risks. As shown in Fig. 7, the living hinge offered the simplest manufacturing approach, enabling integration of all joints into a single 3d-printed component without assembly.

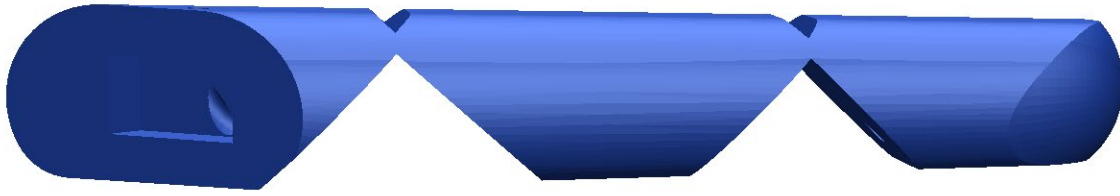


Fig. 7: CAD model of Digit with living hinges

Fatigue resistance of the living hinges proved inadequate, with permanent deformation and cracking after 50 cycles of extension and retraction. This failure rendered the living hinge option unsuitable for a toy application, prompting a transition to revolute joints, shown in Fig. 8. The revolute joints are retained by self-tapping bolts, allowing easy rotation by the joint axis while eliminating the permanent bending strain of the living hinges.

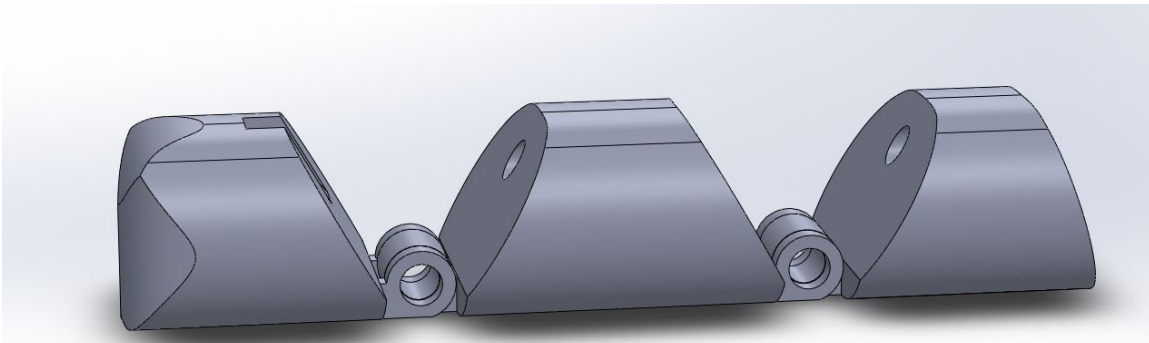


Fig. 8: CAD model of Digit with revolute joints

The hole sizing of the joints was intentionally asymmetrical, with one side being closer to the bolt's outer diameter, while the other side had more clearance. This created a tight hole for the bolt to thread into, while allowing the other thumb segment to rotate freely. Furthermore, Fig. 9 shows the revolute joint repositioned closer to Digit's vertical axis to give space for an elastic band channel to be placed behind the thumb.

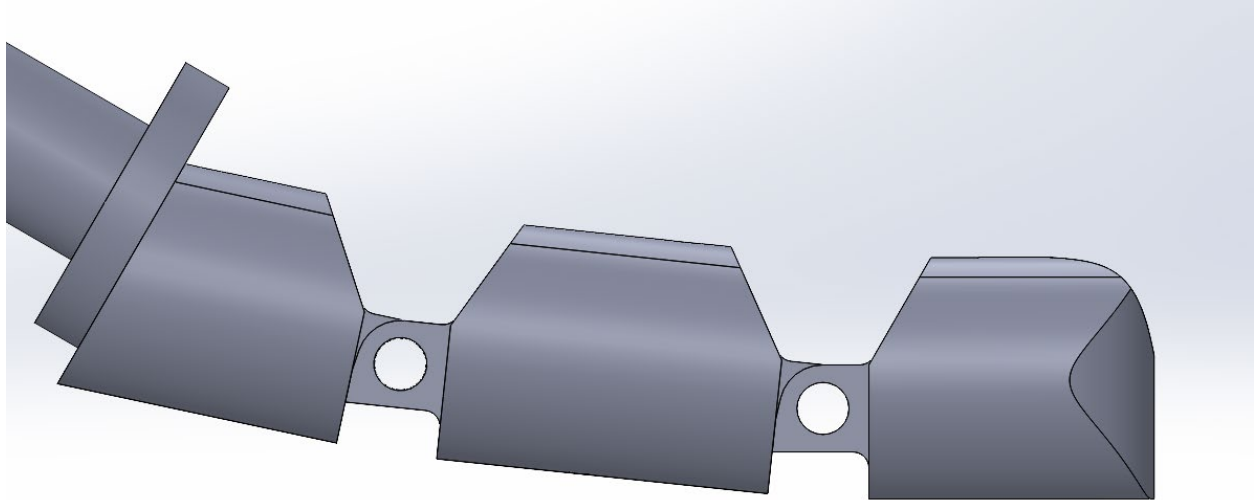


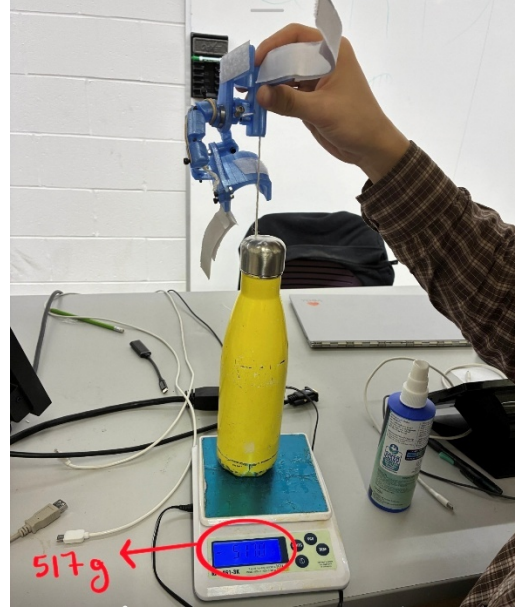
Fig. 9: CAD model of Digit with revolute joints aligned with vertical axis and hard stops

Hard stops were added to the joints where the thumb segments meet. These hard stops take the reverse load, physically preventing the segments from peeling backwards from the elastic band's pulling. Ultimately, the revolute joints showed no cracking and high fatigue resistance under cyclic loading making them the most reliable option.

Further testing showed a significant decrease in input actuation force when using one joint compared to two joints as shown in Fig. 10(a). Testing was done using a scale calibrated to the weight of a water bottle, which was then slowly lifted until the thumb actuated, measuring peak input force. Five tests were conducted for each joint configuration, with the average values showing an 80% decrease in required force. The greater input force in two-joint actuation as demonstrated in Fig. 10(b) is theorized to be from binding issues between the string and the first joint, causing increased resistance to actuate the second joint. Therefore, the second joint was removed due to the substantial difference in input actuation force.



a) Strength test of one-joint configuration



b) Strength test of two-joint configuration

Fig. 10: Strength testing joint configurations

This hinge design satisfies constraints and criteria defined for Digit. It gives each thumb segment free rotation independent of yaw, fully constrains backwards rotation via hard stops, and allows 90 degrees of retraction range of motion. Actuation effort is also low due to the low joint friction.

3.2.2 Yaw Mechanism

The thumb yaw mechanism satisfies the required criteria, with manual testing showing >80 degrees of motion and retraction mechanism independence. The mechanism uses a linkage-based design to convert yawing motion of the wearer's wrist into displacement of a push rod, which mounts off-center to the thumb's axis of rotation. By changing the yaw angle of their wrist, the wearer can change the yaw angle of the thumb relative to their hand. The mechanism was inspired by RC-plane push rods, which use a similar linkage to change the angle of control surfaces. 3D-printing was the primary fabrication method used due to its availability, accuracy, and speed. An x-contact bearing was chosen for the thumb rotation due to the combination of both radial and axial load capability. Velcro strapping was used for mounting, allowing easy adjustment for attaching to different hand and wrist sizes.

As seen in Fig. 8, a set screw was chosen to attach the linkage to the 3D-printed thumb, acting as both the axial retention for the thumb and a mechanism to change the clocking of the thumb relative to the linkage. Fig. 11 also shows the retraction string routing, which critically is collinear with the axis of rotation of the thumb, allowing the thumb to yaw without changing the string-path

length. This means the thumb movements are kinematically independent, resulting in more realistic and intuitive use.

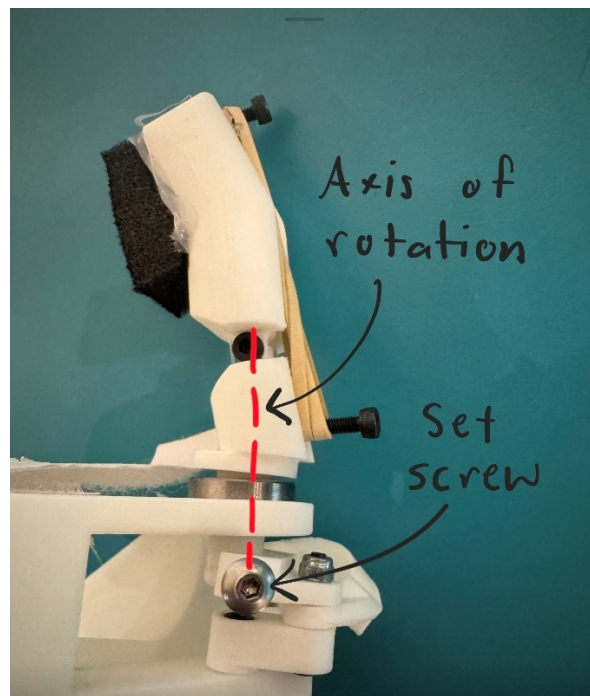


Fig. 11: Close-up of yaw mechanism

Fig. 12 shows the linkage dimensions and synthesis through a SolidWorks sketch, capable of 131 degrees of yawing from wrist motion. Dimensions were optimized to position the linkage as close to the wrist as possible, reducing the overall size of the toy. Linkage lengths and positions were determined using an average adult's wrist yawning range of motions hand dimensions [6]. Adult dimensions were chosen for ease of testing on the design team's hands.

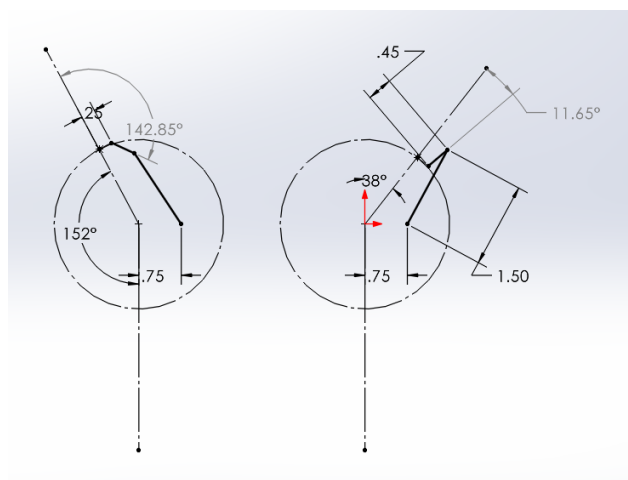


Fig. 12: SolidWorks path length linkage sketch

Fig. 13 shows the pitch axis design of the linkage. To allow the wearer a full range of wrist motion, a joint was added along the pitch axis, aligned with the wearer's wrist pitch axis. This ensures the linkage length changes only during yawing motions, allowing the wearer to pitch their wrist without affecting the thumb, making Digit more intuitive and realistic.

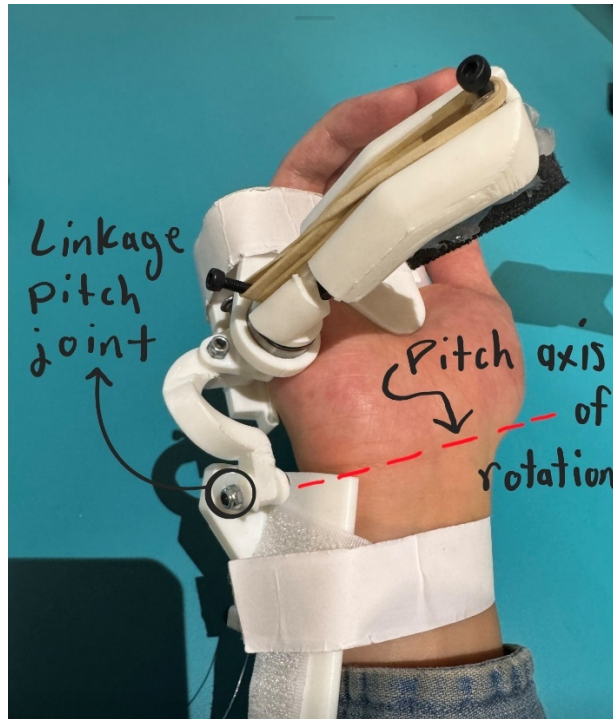


Fig. 13: Pitch axis joint in-line with wrist

The linkage underwent two geometry iterations. As shown in Fig. 14, the first iteration had the links collinear when the wrist was yawed maximally to the left causing the linkage to go over-center and lock up. This issue was resolved by increasing the length of the link closest to the wrist, providing a buffer to prevent going over-center and ensuring smooth motion.

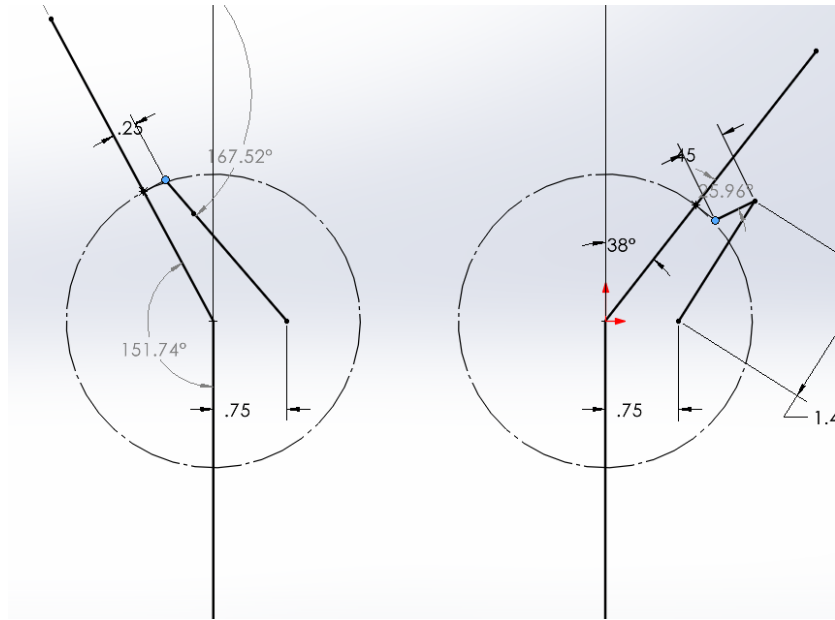


Fig. 14: SolidWorks first iteration of yawing linkage sketch

3.2.3 Retraction Mechanism

Fig. 15 shows the most recent iteration of the retraction mechanism. This was inspired by, "A Body-Powered Underactuated Prosthetic Finger Driven by MCP Joint Motion", which uses cabling attached to the back of the wearer's hand to actuate an additional finger joint when their finger moves [2]. Digit's retraction mechanism uses the pitching motion of the wearer's wrist to actuate the thumb by routing the cable off-axis to the wrist's center of rotation. Pitching the wrist changes the string's path length, pulling the thumb and causing it to retract.

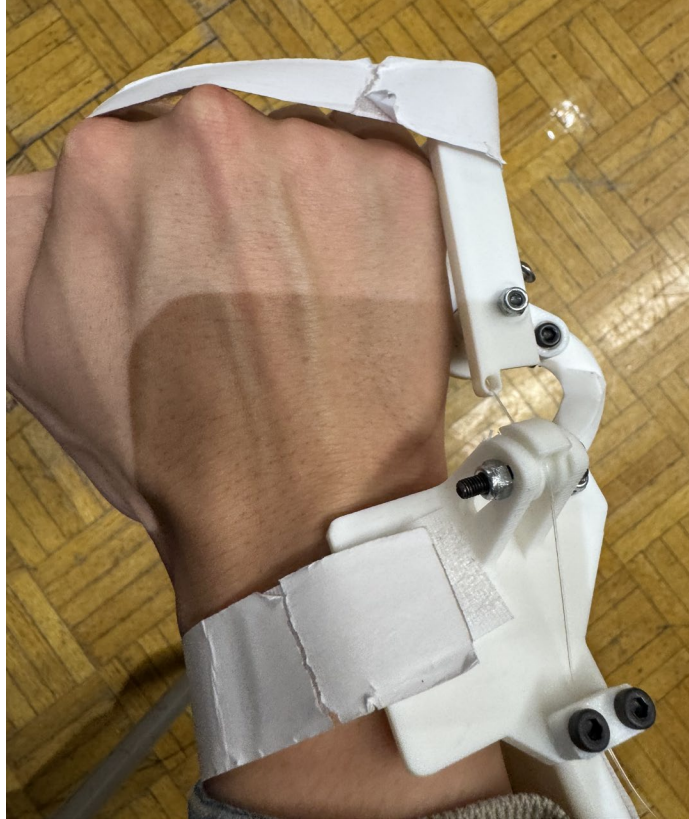
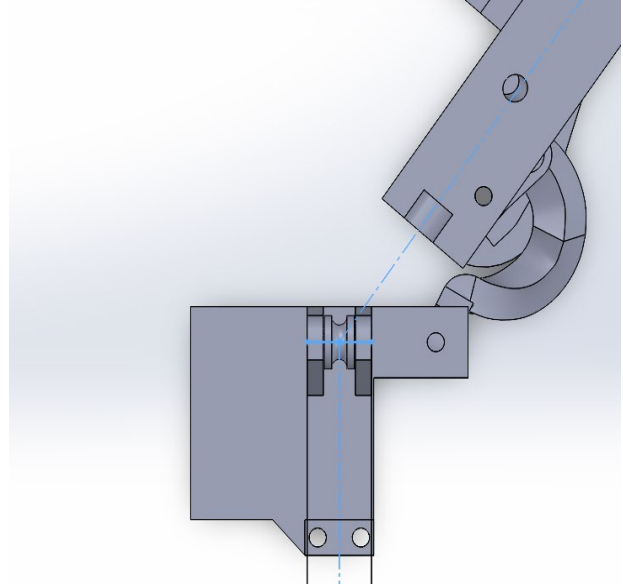
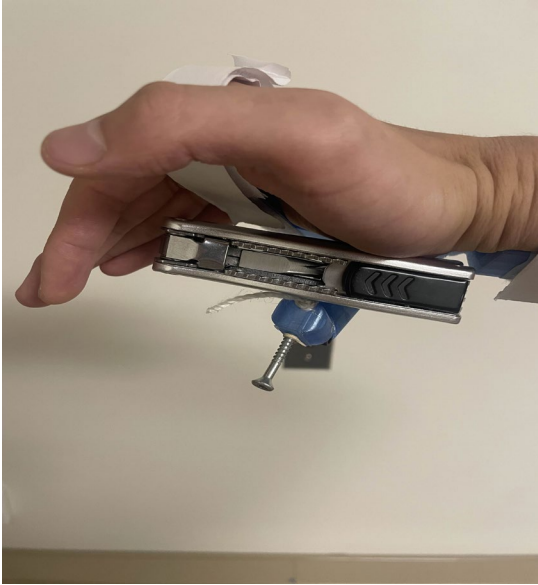


Fig. 15: Wrist-based retraction mechanism

Manual testing showed this mechanism achieved 120 degrees of yawing, satisfying the constraint of >80 degrees of actuation, and Fig. 16(a) shows Digit holding >250 grams. The test used a Leatherman Surge, weighing 350 grams, proving this design far exceeds the required constraints [7].

To accommodate variations in hand and wrist dimensions among users, a movable pulley was incorporated to adjust for changes in path length. The pulley is mounted within a slotted track, providing 1" of adjustability, and secured in position with a clamping bolt. This pulley also acts as a string tensioner, and as shown in Fig. 16(b), it is aligned with the wrist's yaw axis of rotation ensuring that the string path length remains constant during yawing motion.



(a) *Photo of Digit weight holding test*

(b) *SolidWorks assembly with wrist center of rotation*

Fig. 16: Weight-holding test and alignment of pulley with yaw axis of rotation in CAD

The previous iteration of the retraction mechanism used the wearer's thumb to actuate Digit. As shown in Fig. 17, the change in path length of Digit's thumb geometry was designed to closely match that of the wearer's thumb to synchronize motion. However, this dimension varies based on thumb length and mobility, necessitating a path-length adjustment mechanism. This was achieved by placing a pulley off-center from the mechanical thumb's axes of rotation, allowing offset adjustments to modify the change in path length. The mechanism was not fully implemented due to the abandonment of this actuation approach.

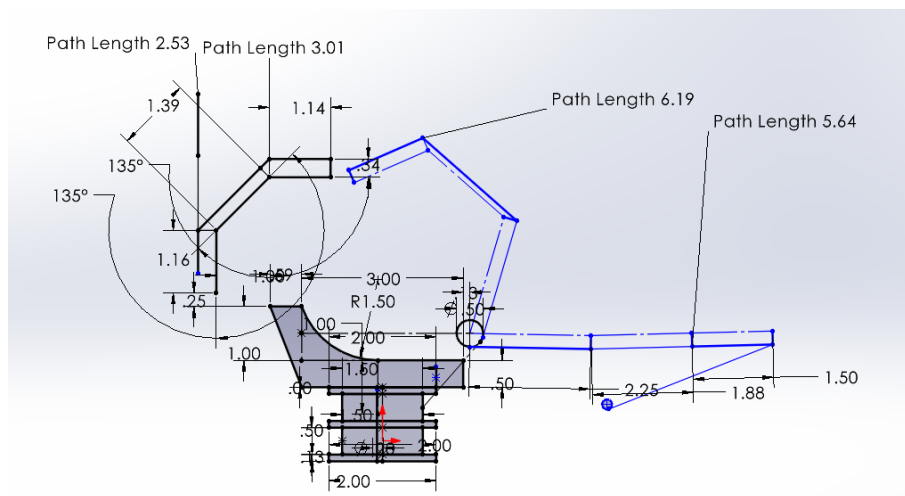


Fig. 18 shows the most recent iteration of the thumb-based actuation design. A string was routed along the back of the hand to the wearer's thumb, where a hook was mounted. Flexion thumb movement caused string path length changes, retracting the added thumb. Thumb-based retraction was rejected due to unintentional actuation, as non-flexion movements also caused string path length changes, making the mechanism unintuitive.



Fig. 18: Demonstration of thumb-based actuation

3.2.4 Extension Mechanism

As seen in Fig. 19, the extension mechanism uses elastic bands to pull the thumb straight. The elastic sits in guide cutouts which are offset outward from the thumb's pivots, causing retraction of the thumb to increase the elastic length, increasing elastic resistance. Manual testing showed this mechanism satisfied >90 degrees of extension and <1 s actuation, with little resistance, so no additional iterations were required.



Fig. 19: Elastic extension mechanism

3.3 Remaining Challenges

Digit exceeds the criteria for a minimum viable product. All mechanisms perform their intended functions while meeting their constraints, but the ease of adjustability could be improved. As seen in Fig. 20, three adjustable mechanisms require the user to loosen bolts. This is not practical for the intended demographic, so a remaining challenge is replacing these fasteners with quick-change mechanisms.

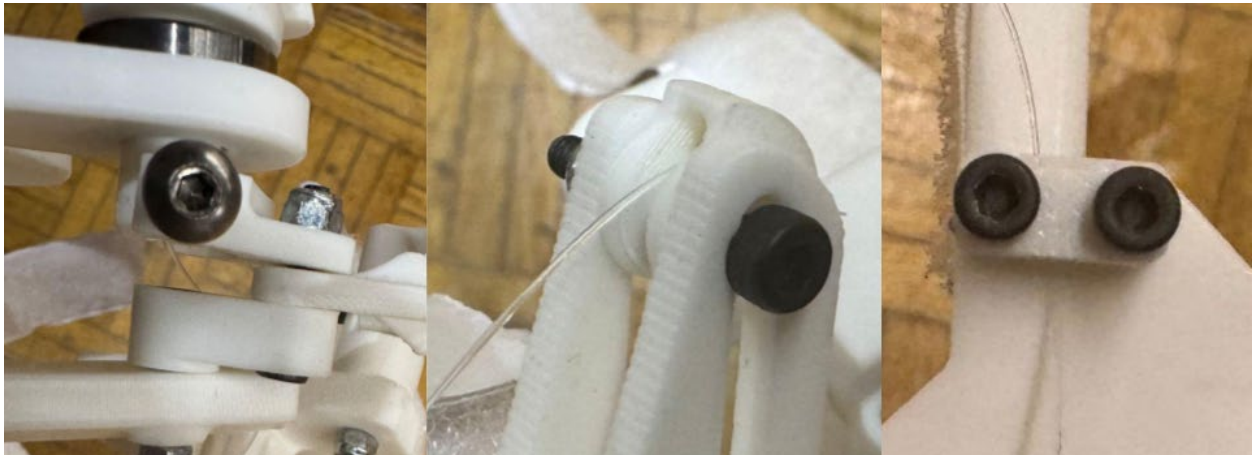


Fig. 20: Adjustable mechanisms on Digit using bolts

4.0 Conclusions and Recommendations

4.1 Conclusion

The current stage of development demonstrates the mechanical viability of Digit, and its potential as a Spin Master product. The prototype successfully meets the functional requirements and constraints defined in Section 2. Digit provides two degrees of freedom, with 80 degrees of yaw and 80 degrees of retraction/extension. Through iterative designs and prototypes described in Section 3, the current version achieves consistent and reliable mechanically powered actuation without the use of electronics. Yawing is controlled through wrist yaw rotation, and retraction/extension through wrist pitch rotation. Despite meeting functional requirements, the current design is not yet fully optimized for practicality: adjustments require loosening and tightening bolts, and the elastic tension force is not adjustable.

4.2 Recommendations

Further development of Digit should focus on addressing the current design limitations to improve consistency, ergonomics, and user accessibility. The current revolute joints that connect the thumb pieces together can be replaced with 3D-printed TPU joints to reduce part count. Furthermore, the current method of relying on bolts for adjusting mechanisms should be replaced with a quick-change mechanism such as ratcheting straps to be better suited for children. Additionally, implementing a mechanism to change elastic tension could further improve the ergonomics of Digit. Together, these changes would significantly improve comfort and accessibility, while maintaining the fundamental mechanical architecture of Digit.

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